MILITARY SPECIFICATION

POWER SUPPLY, TELETYPEWRITER
PP-1767( )/UG

1. SCOPE

1.1 This specification covers requirements for a power supply unit capable of delivering regulated and adjustable direct current for high level (115-120V) teletypewriter signal circuits.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-Q-9858	- Quality Program Requirements
MIL-M-15071	- Manuals, Equipment and Systems
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore; General Specification
MIL-E-17555	- Electronic and Electrical Equipment, Accessories and Repair Parts; Packaging and Packing of
MIL-R-22732	- Reliability Requirements for Shipboard and Ground Electronic Equipment

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-189	- Racks, Electrical Equipment, 19-inch and Associated Panels
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-461	- Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-701	- List of Standard Semiconductor Devices
MIL-STD-781	- Reliability Tests: Exponential Distribution
MIL-STD-785	- Requirements for Reliability Program (for Systems and Equipments)

PUBLICATION

MILITARY

MIL-HDBK-217	- Reliability Stress and Failure Rate Data for Electronic Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article sample. - Prior to beginning production, a sample shall be tested as specified in 4.3.

3.2 Design objectives. - The equipment design shall be such as to attain the basic objectives of MIL-E-16400.

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3.3 Modular construction provisions of MIL-E-16400 are not applicable.

3.4 Design requirements. -

3.4.1 General design. - The unit shall consist basically of solid state rectifiers with regulating and control circuits to achieve the performance specified in 3.4.4.

3.4.2 Material and parts. - The material and parts used in the construction of this equipment shall be selected in accordance with MIL-E-16400. Rectifier and regulator elements shall be solid state, selected in accordance with MIL-STD-701. The contractor may elect to use nonstandard parts as described in paragraph titled "Nonstandard Parts" of MIL-E-16400. Items or processes which require approval of the procuring activity are listed in 6.3 of MIL-E-16400.

3.4.3 Environmental conditions. - The equipment shall meet the requirements of paragraph titled "Temperature conditions" (class 4 of table II) of MIL-E-16400. The equipment shall also meet the requirements of paragraph titled "Humidity" of MIL-E-16400.

3.4.3.1 Thermal design shall be in accordance with paragraph titled "Thermal design" of MIL-E-16400.

3.4.4 Electrical design. -

3.4.4.1 Primary power will be single phase 120 Vac ±10 percent at 50 or 60 Hz ±5 percent.

3.4.4.2 The power supply shall provide four outputs:

(a) 6 Vac ±5 percent at 1.5 ampere, unregulated
(b) 120 Vdc at 1 ampere, adjustable
(c) 115 Vdc at no load, maximum 0.06 ampere, regulated
(d) 115 Vdc at no load, maximum 0.06 ampere, regulated

3.4.4.2.1 A locking type screw driver adjustment shall be provided to permit manual adjustment of the 120 Vdc output voltage for the pull range of rated line and load variations.

3.4.4.2.2 Each of the two 115 Vdc signal outputs shall provide nominal voltage ±5 percent at no load when the 120 Vdc output is adjusted for its nominal voltage. A potentiometer control shall be provided to adjust each signal current for 60 milliamperes. When so adjusted, the outputs shall become constant-current sources, within ±2 percent for the full range of rated line and load variations.

3.4.4.2.3 Ripple in the dc outputs shall not exceed 1 percent rms when working into a resistive load between zero and rated maximum output.

3.4.4.3 Output circuits shall be designed to permit grounding of either positive or negative terminals.

3.4.4.4 There shall be a fuse, with blown fuse indicator, in each lead of the input circuit and the dc output circuits. There shall be protection of the output circuits in case of a fault in the equipment which might draw power from a dc line.

3.4.4.5 A power input switch shall be provided which will completely disconnect the ac input from the equipment when in the "OFF" position.

3.4.4.6 The power supplies shall be provided with a suitable ammeter for monitoring the 120 Vdc output current, or the current in either of the 115 Vdc signal circuits. The scale shall read 0 to 1 ampere in the 120 Vdc line, 0 to 100 mA in the signal circuits. Switching the meter for various readings shall cause no appreciable disturbance of the outputs. The meter switch shall have an "OFF" position.

3.4.4.7 A chassis ground terminal shall be provided.

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VARIABLE TRANSFORMER (SEE TABLE III)

3.4.5 Mechanical design. - Mechanical design shall be in accordance with paragraph titled "Mechanical Design" of MIL-E-16400, and applicable subparagraphs thereunder.

3.4.5.1 Enclosure. - The power supply shall be packaged as a self-contained unit, with provision for adaptor brackets to mount in a standard 19-inch relay rack in accordance with MIL-STD-189. It shall have a suitable cover with convenient access for inspection and replacement of components. Degree of enclosure shall be as defined for "Protected" in MIL-STD-108.

3.4.5.2 Size and weight. - Width shall be such as to permit mounting in accordance with 3.3.5.1. Depth shall not exceed 7 inches, not including handles. Height shall be standard 7 inches. Weight shall be minimum necessary to achieve the specified performance, reliability, and maintainability, not to exceed 45 pounds.

3.4.5.3 Connection to the power units shall be made to screw-type barrier terminals accessible by a hinged cover at the top of the unit. The terminal boards shall have insulated protective covers or other measures to minimize the hazard of exposed connections.

3.4.5.4 Front panel. - All controls, indicators, fuses, and switches shall be mounted on the front panel, except the potentiometer for setting the 120 Vdc output, which shall be accessible adjacent to the terminals at the top of the unit. Suitable handles shall be provided on the front panel to facilitate installation and maintenance of the equipment. Locking devices shall be provided on all controls in accordance with paragraph titled "Locking devices" of MIL-E-16400.

3.4.5.4.1 Panel layout shall be submitted for approval in accordance with paragraph titled "Panel layouts and arrangement of operating controls" of MIL-E-16400.

3.4.5.4.2 A power-on indicator light with a red jewel shall be mounted on the panel.

3.4.5.5 Painting and treatments shall be in accordance with paragraph titled "Coating and treatments" of MIL-E-16400 and applicable subparagraphs thereunder.

3.4.5.6 Test points. - Significant voltages, currents, and wave forms shall be measurable at test points to facilitate alignment and determination of circuit faults, in accordance with paragraph titled "Test points and test features" of MIL-E-16400.

3.5 Radio interference and undesired radiation. - The equipment shall meet the radio interference and undesired radiation requirements of MIL-STD-461 for class IC equipments.

3.6 Reliability.-

3.6.1 Quantitative reliability requirement.- The minimum acceptable MTBF (Q_1 as defined by MIL-STD-781) of this equipment shall be 10,000 hours.

3.6.2 Reliability program plan.- The contractor shall submit a reliability program plan in accordance with and in paragraph sequency with MIL-STD-785. Particular attention should be paid to those sections of MIL-STD-785 which require submission of detailed tasks, procedures to implement and control these tasks, and time schedule. The proposed reliability program plan shall be submitted as a separate and complete entity within the total system or equipment/project delineation. The reliability program plan as approved and incorporated into the contract becomes the basis for contractual compliance. The plan, including reliability test plans, shall describe an integrated effort within the total project; it shall provide information to show how the contractor will meet specified quantitative reliability requirements during development and manufacture. The final reliability program plan shall be submitted for Government approval.

3.6.3 Reliability prediction.- The contractor shall submit a reliability prediction in accordance with paragraph 5.0 of MIL-HDBK-217 to the procuring activity. The prediction shall assume a 50° C ambient temperature and a shore environment. The contractor shall evaluate the prediction with respect to the reliability expected from a production unit. If there is a discrepancy between the predicted MTBF and required MTBF the contractor shall identify the causes of the discrepancies and indicate action required to assure that the

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inherent reliability of the equipment is preserved during the production process. The processes and procedures required to assure production of reliable equipment shall be thoroughly documented.

3.6.4 Source of failure rates.- The primary source of failure rate data shall be MIL-HDBK-217. For items not included in MIL-HDBK-217, other sources of data may be used with the approval of the procuring activity. When such alternate sources of data are proposed, enough information shall be submitted to enable the procuring activity to evaluate the validity and applicability of the results derived therefrom.

3.6.5 Reliability reports.- Reports providing a complete accounting of progress on each element defined by the reliability program plan, results achieved, and status of actions to resolve problems and correct weak links shall be submitted monthly to the procuring activity.

3.7 The equipment shall conform to the paragraph titled "Supplementary requirements" of MIL-E-16400, as applicable.

3.8 Manuals.- When required by contract or order, instruction manuals shall be type I in accordance with MIL-M-15071.

3.9 Safety.- Requirement 1 of MIL-STD-454 shall apply to this power supply unit.

3.10 Workmanship.- Workmanship shall be in accordance with Requirement 9 of MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system.- The contractor shall provide and maintain an effective inspection and quality control system acceptable to the Government covering the supplies under the contract. The quality control system shall be in accordance with MIL-Q-9858. A current written description of the system shall be submitted to the command or agency concerned, via the cognizant Government inspector, for system approval, 60 days prior to first article inspection. If first article inspection is not required, the written description of the quality control system shall be submitted for approval 60 days prior to production inspection. (An advance copy shall be forwarded to the command or agency concerned.) The written description shall include flow charts showing the complete flow of material from inspection on receipt, through all manufacturing processes, to final shipment. The various processes and inspection positions shall be clearly identified on the flow charts and cross referenced to the detailed inspection procedures and criteria of the written description of the quality control system. Any changes to the approved quality control system which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the command or agency concerned for approval prior to use.

4.1.2 Government verification.- All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practice, methods, and procedures of the written system description are being properly applied (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all the requirements of this specification. Failure of the contractor to correct promptly deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Failure reporting.- Failure reporting shall be in accordance with MIL-E-16400.

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4.1.4 Tests, test equipment, and test procedures.- The contractor shall submit a list of proposed tests and a copy of the written test procedures, including test equipment and tolerance limits to be used in testing first article and production equipment to the command or agency concerned, via the Government inspector, for review and approval, 60 days prior to beginning first article tests. An advance copy shall be forwarded to the command or agency concerned. Each test shall be identified in accordance with 4.2 as to classification. The contractor may, at his option, submit the production portion 60 days prior to beginning production. Testing shall not begin until approval is received.

4.2 Classification of inspection.- The method of examination and testing the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection
    (1) Production inspection
    (2) Production control inspection
    (3) Environmental inspection

4.3 First article inspection.- First article inspection shall consist of all examination and testing necessary to determine compliance with this specification. First article inspection shall include the inspections specified in table I.

TABLE I. First article inspection

Inspection	Reference
Surface examination	4.5.1
Operating test	4.5.2
Weights and dimensions	MIL-E-16400
Supply line voltage and frequency	MIL-E-16400
Heat test	MIL-E-16400
Temperature tests	MIL-E-16400
Humidity test (see 4.3.1)	MIL-E-16400
Power	MIL-E-16400
Controls and control circuits	MIL-E-16400
Accelerated life test	MIL-E-16400
Shock and vibration	MIL-E-16400
Reliability	Test level No. 2, MIL-R-22732 shall be used for determining environment for reliability testing.
Electromagnetic interference characteristics	MIL-STD-461, class 1C

4.3.1 Humidity tests required herein shall include supply line voltage and frequency tests to verify compliance with the detailed requirements under 3.4.2, 3.4.2.1, 3.4.2.2, and 3.4.2.3, herein.

4.3.2 First article inspection report.- The first article inspection report shall be submitted within 30 days of completion of the first article to the command or agency concerned, via the Government inspector, for review and approval. An advance copy shall be forwarded to the command or agency concerned. The first article inspection report shall contain results of all tests and shall include completed Form NAVEXOS 3373 for shock test and, where resilient mounts are employed, compliance with figure 3 of MIL-E-16400. First article approval will not be granted unless the report is complete. Any production entered into prior to approval of the first article report will be at the supplier's risk.

4.3.3 First article model acceptance.- Acceptance of the first article model shall be contingent upon approval by the command or agency concerned with the first article inspection report. Any further laboratory testing of the first article model may be made at the discretion of the Government.

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4.4 Quality conformance inspection.-

4.4.1 Production inspection.- Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with 4.1.4 and shall include the following:

Surface examination	(4.5.1)
Operating test	(4.5.2)

4.4.2 Production control inspection.- Production control inspection, including sampling, shall conform to table II and to the procedures for special inspection levels of MIL-STD-105. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE II. Production control inspection

Inspection	Reference paragraph	AQL percent
Weights and dimensions	MIL-E-16400	6.5
Supply line voltage and frequency	MIL-E-16400	6.5
Heat test	MIL-E-16400	6.5
Power	MIL-E-16400	6.5
Electromagnetic interference	MIL-E-16400	6.5
Controls and control circuits	MIL-E-16400	6.5

4.4.2.1 Rejected lots.- If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection.- Environmental inspection shall be conducted to prove the durability of the equipment to withstand the effects of extremes of environment, service conditions, and hard usage. Environmental inspection shall be performed on equipment or equipments that have passed production inspection in accordance with 4.4.1 and shall include the tests listed in table III.

TABLE III. Environmental inspection

Inspection	Reference paragraph
Temperature	MIL-E-16400
Humidity (see 4.3.1)	MIL-E-16400
Accelerated life test	MIL-E-16400
Shock and vibration	MIL-E-16400

4.4.3.1 Environmental inspection samples.- Samples for the environmental inspection shall be selected without regard to their quality in accordance with the following:

(a) One sample from the [illegible]
(b) One sample from each successive 100 units produced or fraction thereof

4.4.4 Nonconformance and retest.- If a sample unit fails a test specified in 4.4.2 or 4.4.3, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector the results

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thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, and processes. The equipment shall then be retested.

4.4.5 Reinspection of conforming production control and environmental test units.- Unless otherwise specified, sample units which have been subjected to and passed production control and environmental inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all visible damage.

4.4.6 Quality conformance inspection report.- The contractor shall submit to the Government inspecting activity prior to shipment of the respective equipment a complete report detailing the results of all testing performed. The report shall list the serial numbers of equipments tested during the reporting period. During the conduct of tests, performance parameters shall be measured and verified in accordance with this specification. The report shall include completed Form NAVEXOS 3373 for the shock test and, when resilient mounts are used, compliance with figure 3 of MIL-E-16400. Equipment will not be accepted unless the report is complete.

4.5 Inspection procedures.-

4.5.1 Surface examination.- Equipment shall be examined for the following:

(a) Workmanship, assembly and fit, mechanical safety, and marking
(b) Materials, parts, coatings, and finish

4.5.2 Operating test.- The equipment shall be energized and subjected to an operating test to insure qualitatively the proper functioning of the equipment, including all operating controls and conformance with the safety requirements. This test shall include, but not be limited to, supply line and frequency tests to verify compliance with the applicable detailed requirements specified herein.

4.5.3 All other inspections referenced herein shall be conducted in accordance with the paragraph titled "Inspection procedures" of MIL-E-16400, as applicable.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking.- Unless otherwise specified in the contract or order, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use.- The equipment covered by this specification is intended for service in ship and shore stations where it is expected to withstand continuous use for long periods, under military service conditions, without benefit of overhaul. The equipment is in each case a vital instrument; failure at a critical moment invariably results in serious reduction in the battle efficiency. Emergency repairs can seldom be made with sufficient celerity to avoid having equipment out of service during a critical period.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of preservation, packaging, packing, and marking (see 5.1).

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6.3 First article.- Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy-EC

(Project 5815-N048)

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